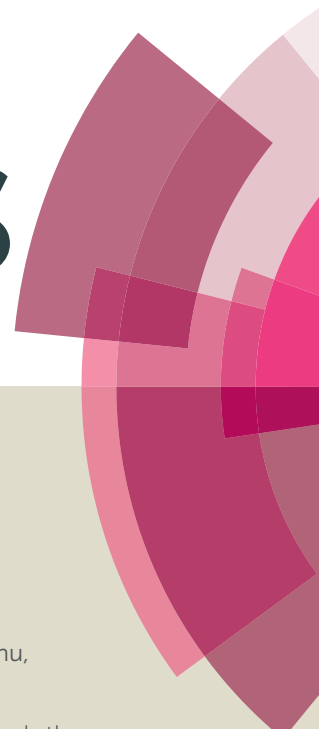


RSC Advances



This article can be cited before page numbers have been issued, to do this please use: J. Xu, J. Li, Y. Zhu, K. Zhu, Y. Liu and J. Liu, *RSC Adv.*, 2016, DOI: 10.1039/C6RA02865A.



This is an *Accepted Manuscript*, which has been through the Royal Society of Chemistry peer review process and has been accepted for publication.

Accepted Manuscripts are published online shortly after acceptance, before technical editing, formatting and proof reading. Using this free service, authors can make their results available to the community, in citable form, before we publish the edited article. This *Accepted Manuscript* will be replaced by the edited, formatted and paginated article as soon as this is available.

You can find more information about *Accepted Manuscripts* in the [Information for Authors](#).

Please note that technical editing may introduce minor changes to the text and/or graphics, which may alter content. The journal's standard [Terms & Conditions](#) and the [Ethical guidelines](#) still apply. In no event shall the Royal Society of Chemistry be held responsible for any errors or omissions in this *Accepted Manuscript* or any consequences arising from the use of any information it contains.



Journal Name

ARTICLE

A triPEG-boron based electrolyte membrane for wide temperature lithium ion battery

Junyi Xu, Jie Li, Yuewu Zhu, Kai Zhu, Yexiang Liu, Jin Liu*

Received 00th January 20xx,
Accepted 00th January 20xx

DOI: 10.1039/x0xx00000x

www.rsc.org/

A triPEG-boron based solid polymer electrolyte (SPE) membrane is fabricated for all-solid-state lithium ion battery. With an ether oxygen (EO):LiN(SO₂CF₃)₂ (LiTFSI) molar ratio of 25:1, this electrolyte membrane has ionic conductivities of 2.95×10^{-5} S cm⁻¹ and 1.28×10^{-3} S cm⁻¹, and electrochemical decomposition potentials of 5.23 V and 4.88 V, respectively at 25 °C and 120 °C. The all-solid-state LiFePO₄/SPE/Li batteries show high cycling performance in the temperature range of 25–140 °C. At 25 °C and 0.3 C, a discharge capacity of 84 mAh g⁻¹ is obtained at the 200th charge-discharge cycle with approximately 100% coulombic efficiency. The battery also works well at 140 °C and 3 C, delivering a discharge capacity of 146 mAh g⁻¹ initially and 113 mAh g⁻¹ after 200 cycles. These results show that this kind of all-solid-state lithium ion battery has a wider working temperature range than conventional liquid electrolyte lithium ion batteries, indicating its suitability for use in the fields of high safety requirements.

1. Introduction

Solid polymer electrolyte (SPE) has received considerable attention as a key component for all-solid-state lithium ion battery.^{1–3} Through the use of SPE, safety risks in conventional liquid electrolyte lithium ion batteries can be significantly reduced, especially at high operating temperatures of greater than 50 °C.^{4–6} Currently, poly(ethylene oxide) (PEO) based electrolyte is the most popular SPE used in all-solid-state lithium ion battery.^{7–9} However, this kind of battery works only at temperatures greater than 60 °C due to low room temperature ionic conductivity of the electrolyte.^{10,11} Considerable research effort has been dedicated to improving SPE performance, such as copolymerizing and plasticizing with low molecular weight organics.^{12–18} But, high temperature durability of the SPE is destroyed after such modifications. For example, the LiFePO₄/Li battery using the polysiloxane based SPE with a room temperature ionic conductivity of 7.96×10^{-5} S cm⁻¹, delivered a capacity of 144 mAh g⁻¹ at 25 °C.¹⁸ But, obvious capacity decay occurred at 80 °C. In addition, despite the improvements in high temperature properties offered by modifications of adding nanoparticles to PEO based electrolytes, the room temperature performance was depressed.^{19–21} Thus, it is crucial to develop SPE that maintains both high ionic conductivity and robust thermal stability to enable a wider operating temperature range for all-solid-state lithium ion battery.^{22,23}

It is known that the boron atom is sp²-hybridized and

contains an empty p-orbital that can interact with basic anion of lithium salt.^{24,25} Previous studies indicated that a small moiety of boron compounds as chemical additives could increase the lithium ion transport ability of electrolytes.^{26–30} But to date, the ability of boron to remodel molecular structure for improving other properties of the SPE has not been reported. In this study, we develop a triPEG-boron based SPE membrane for all-solid-state lithium ion battery. With boron linked, three PEG molecules constitute a Y-shape structure to substitute for the linear structure. As a result, the ionic conductivity, film-forming property and elevated temperature properties of the electrolyte are enhanced. The all-solid-state LiFePO₄/Li battery assembled with the electrolyte displays high cycling performance in the wide temperature range of 25–140 °C.

2. Experimental

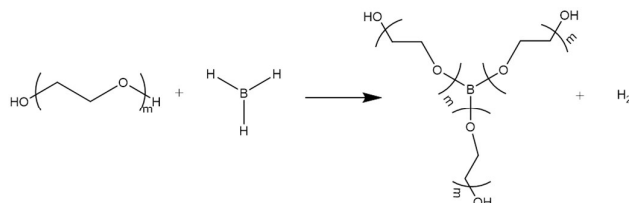
2.1 Materials

Poly(ethylene glycol) (PEG, Mw = 570–630, liquid) were purchased from Aladdin and further purified by vacuum drying at 80 °C for 24 h to remove residual moisture. Borane-tetrahydrofuran (BH₃/THF, 1 mol L⁻¹ in THF) and lithium bis(trifluoromethylsulfonylethyl)imide (LiTFSI > 99%) were purchased from J&K company and used as received. Acetonitrile (HPLC, 99.9%) and N-methyl-2-pyrrolidone (NMP, HPLC, 99.5%) were provided by Tianjin reagent company. Poly(vinylidene fluoride) (PVDF, Mw = 700,000) was provided by ARKEMA and dried at 60 °C before use.

School of Metallurgy and Environment, Central South University, Changsha City, 410083, China. E-mail: jinliu@csu.edu.cn

2.2 Synthesis of B-PEG

The triPEG-boron polymer (B-PEG) was synthesized via the reaction between PEG and BH_3/THF without any catalyst. The PEG was first dissolved in acetonitrile and then the BH_3/THF solution was added dropwise by a constant pressure titration funnel. The reaction was conducted at 45 °C for 18 hours.



Scheme 1 Chemical reaction for the formation of B-PEG.

The molar ratio of the BH_3 and the PEG was maintained at 2:3, and the synthesis was carried out under argon atmosphere. The excess of BH_3 was added to realize a complete reaction between PEG and BH_3 . Then, the redundant BH_3 was removed by heating the sample at 100 °C for 24 h and B-PEG was obtained. The B-PEG polymer is solid, while compared to the original PEG is just liquid, demonstrating the synthesis. This reaction may follow the equation as shown in Scheme 1.

2.3 Preparation of SPEs

The triPEG-boron based electrolyte membrane was prepared by solution casting method. For the electrolyte B-PEG/LiTFSI, 79.3 wt% of polymer B-PEG and 20.7 wt% of LiTFSI were dissolved in acetonitrile, and then the solution was poured into a Teflon mould. After the solvent was removed from the electrolyte at 120 °C for 24 hours in argon atmosphere, a colorless and transparent electrolyte membrane of B-PEG/LiTFSI was obtained. For B-PEG/PVDF/LiTFSI, similar preparation process was conducted, except that 71.4 wt% of B-PEG, 18.6 wt% of LiTFSI and 10.0 wt% of PVDF^{15,18} were dissolved in N-methyl pyrrolidone with optimized ratio and solvent.

2.4 Characterizations

Fourier transform infrared spectroscopy (FTIR, Nicolet 6700, USA), and ^{11}B -nuclear magnetic resonance (Bruker ASCENDTM 400, German) were used to evaluate the chemical structure. The molecular weight of B-PEG was characterized by gel permeation chromatography (GPC, Waters 1515, USA) to verify its purity. To prepare the test sample, B-PEG was dissolved in tetrahydrofuran according the concentration of 3 wt% in an argon protected environment. Thermal stabilities of B-PEG/PVDF/LiTFSI were examined by thermogravimetry-differential thermal analysis (TG-DTA, SDTQ-600, TA, USA). The samples (about 10 mg) were heated from room temperature to 550 °C under nitrogen atmosphere at a heating rate of 10 °C min^{-1} . The glass transition temperature T_g was examined by differential scanning calorimeter (DSC) from -80 °C to 40 °C under nitrogen atmosphere at a heating rate of 10 °C min^{-1} .

The ionic conductivities (σ) of the electrolyte were determined by AC impedance spectroscopy measurements in

the temperature range of 20 to 130 °C. The electrolyte wafer samples were sealed into 2025 button cells with two stainless steel wafers as blocking electrodes. The sweep frequency ranged from 700 kHz to 10 Hz with AC amplitude of 10 mV applied. The test temperature was controlled by a high-low temperature chamber (SDH4005P, HAOYUAN CO., China). All values of σ were calculated from the following equation:

$$\sigma = l/SR_b$$

where l refers to the thickness of the electrolyte, S refers to the contact area between electrode and electrolyte, and R_b refers to the bulk resistance of the electrolyte, which can be obtained from the AC impedance spectroscopy.

Linear sweep voltammetry tests were applied to the electrolyte membranes to determine the decomposition voltage, with stainless steel as the working electrode and lithium metal as counter and reference electrode. All measurements were conducted at a scan rate of 10 mV s^{-1} from 2 to 6 V vs. Li/Li^+ .

The B-PEG/PVDF/LiTFSI electrolyte membrane was made into wafers with a diameter of 20 mm and a thickness of about 0.5 mm, and then assembled into $\text{LiFePO}_4/\text{Li}$ half cells (CR2025) to conduct galvanostatic cycle performance tests (CT 2001A, LAND, China). The test voltage range was 2.5 to 4.2 V vs. Li/Li^+ when the test temperature was between 25 to 120 °C. For the cycle test at 140 °C, the voltage range was 2.5 to 4.0 V vs. Li/Li^+ . Before the measurements were conducted, cells were pre-warmed at the testing temperature for 0.5 h to ensure a constant test temperature. The resistances of the $\text{LiFePO}_4/\text{Li}$ half cells were characterized by AC impedance test in the frequency range of 2 MHz to 10 mHz with an AC amplitude of 10 mV applied. All the AC impedance and linear sweep voltammetry tests were conducted by an electrochemical workstation (PARSTAT 2273, Princeton Applied Research, USA).

3. Results and Discussion

3.1 Structure characterization of B-PEG

Fig. 1a shows the FTIR spectra of the synthesized product B-PEG (solid) and raw materials of PEG (liquid). In the spectrum of B-PEG, the peak at 669 cm^{-1} is the bending stretch of B-O.³¹ The peak at 1417 cm^{-1} and the shoulder at 1326 cm^{-1} belong to the anti-symmetric stretching vibration of B-O,³¹ illustrating the formation of the boric ester bonds that is in good agreement with previous reports.^{27,32}

The broad peak at 1112 cm^{-1} and the peaks at 1200 - 1500 cm^{-1} which appear both in the spectra of B-PEG and PEG can be assigned to C-O-C from the PEG chains. A strong peak centred at 2850 cm^{-1} is assigned to the CH_2 symmetric stretch that indicates gauche defects in the PEG chains, which would be beneficial in lessening the crystallinity of B-PEG and increasing ionic conductivity of the electrolyte.³³ In addition, the broad peak at the centre of 3354 cm^{-1} is mainly due to the terminal hydroxyl groups in PEG and B-PEG.

In order to analyze the chemical structure of B-PEG, the ^{11}B NMR was conducted and the results are shown in Fig. 1b. Only

two peaks are present in the spectrum. The main peak at 18.41 ppm is attributed to the product of B-PEG.^{34,35} The small shoulder at 1.52 ppm is considered to be from the trace oxidative reagent of BH₃ which was from the sampling process of BH₃/THF. Hence, both FTIR and NMR results reveal that the PEG is linked to the boron atoms.

The molecular weight distribution of B-PEG was measured by GPC, and the results are shown in Fig. 1c. The average

molecular weight (Mw) of B-PEG is 1827. As the Mw of PEG raw materials used is about 600, the chemical structure of B-PEG can be deduced as described in Scheme 1. Every three PEG molecules are linked together by a boron ester group, forming a B-centred structure. The molecular weight peak (Mp) of B-PEG is 1926 and the polydispersity is about 1.15, illustrating that the yield of this reaction is almost 100%.

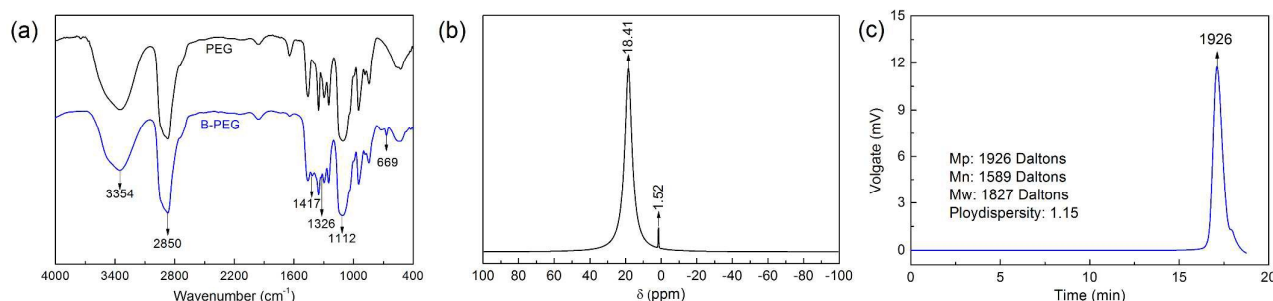


Figure 1 (a) FTIR spectra of PEG and B-PEG. (b) ¹¹B NMR spectrum of B-PEG. (c) The GPC result of B-PEG.

3.2 Electrochemical properties and thermal stability of the triPEG-boron based electrolyte membrane

The molar ratio ([EO]:[Li⁺]) of ether oxygen (EO) in PEG and lithium ions in LiTFSI was optimized, as it is an important influencing factor of ionic conductivity for solid polymer electrolyte. Fig. 2a shows the ionic conductivities of the electrolyte membranes at the [EO]:[Li⁺] ratios of 10, 15, 20, 25, 35 and 50 as a function of temperature. In the temperature range from 20 to 50 °C, the highest ionic conductivity exists when the [EO]:[Li⁺] ratio is 25:1. In the range of 50 to 90 °C, the maximum conductivity gradually migrates to the lower [EO]:[Li⁺] ratio. At 90 °C, the [EO]:[Li⁺] ratio of 10 corresponds to the highest ionic conductivity.

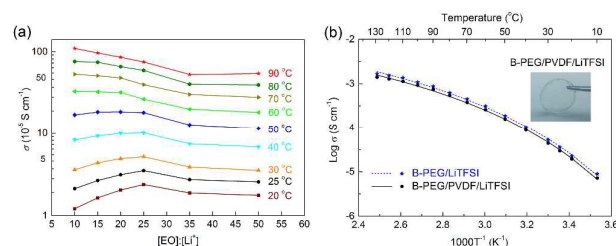


Figure 2 (a) Lithium ion conductivities of the electrolyte membranes of B-PEG/LiTFSI as a function of [EO]:[Li⁺] ratios in the temperature range of 20 - 90 °C. (b) Variable temperature conductivities of B-PEG/LiTFSI and B-PEG/PVDF/LiTFSI. Inset: Photo of B-PEG/PVDF/LiTFSI electrolyte membrane. The solid points are the experimental data, and the solid and short dash lines represent the VTF fitting curves for B-PEG based electrolytes.

This dissimilarity is mainly due to an increase of the segment mobility when the temperature is increased. Higher lithium salt concentration can provide more conduction mediums,^{36,37} but reduce the segmental motion ability.³⁸ The ionic conductivity is decreased as lithium salt increases, thus the higher conductivity is observed in the lower [EO]:[Li] ratio.

At 25 °C, the ionic conductivity is 2.12×10^{-5} , 2.66×10^{-5} , 3.13×10^{-5} , 3.51×10^{-5} , 2.73×10^{-5} and 2.56×10^{-5} S cm⁻¹ when the [EO]:[Li⁺] ratio is respectively 10:1, 15:1, 20:1, 25:1, 35:1 and 50:1. Since room temperature ionic conductivity is crucial for solid polymer electrolyte, the lithium salt content in the following experiments is fixed in the ratio of 25:1.

To improve flexibility and mechanical strength of B-PEG/LiTFSI electrolyte membranes, 10 wt% of PVDF was added.^{15,18} In Fig. 2b the ionic conductivities of the electrolyte membranes without and with PVDF were measured, respectively labeled as B-PEG/LiTFSI and B-PEG/PVDF/LiTFSI. The ionic conductivities are 3.55×10^{-5} , 6.48×10^{-4} and 1.53×10^{-3} S cm⁻¹ for B-PEG/LiTFSI, and 2.95×10^{-5} , 5.44×10^{-4} and 1.28×10^{-3} S cm⁻¹ for B-PEG/PVDF/LiTFSI at 25, 80 and 120 °C, respectively. The conductivity of the B-PEG/PVDF/LiTFSI electrolyte is decreased about 16% on average, but the filming property and flexibility are obviously improved.

The relationship of log σ versus 1000/T of B-PEG/LiTFSI and B-PEG/PVDF/LiTFSI are nonlinear and conform to the Vogel-Tamman-Fulcher (VTF) equation³⁹

$$\sigma = AT^{-1/2} \exp[-B/(T - T_0)]$$

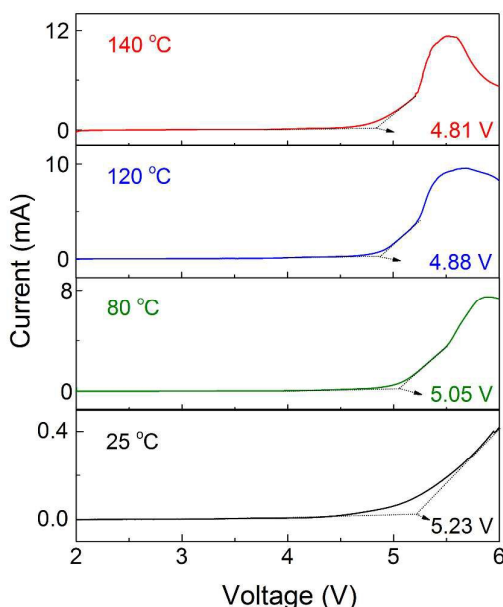
which indicates that lithium ion migration is driven by the segmental motion,⁴⁰ where *A* is the pre-exponential factor, *B* is closely related to the apparent activation energy *E_a* of ion transport and the numerical relationship can be expressed as $B = E_a/(\kappa_b N_A)$, in which κ_b is Boltzmann constant, and *N_A* is Avogadro's constant, and *T₀* is a factor relating to the ideal transition temperature at which the free volume is zero.

The fittings are applied by 1stOpt software, and the fitting parameters and the *E_a* calculated are listed in Table 1. The *E_a* values of B-PEG/LiTFSI and B-PEG/PVDF/LiTFSI are almost equal, illustrating that the addition of PVDF does not change the freedom of ions transport in the electrolyte.⁴¹

Table 1 The VTF fitting parameters and apparent activation energy (E_a) calculated from the VTF equation for B-PEG/LiTFSI and B-PEG/PVDF/LiTFSI electrolyte membranes.

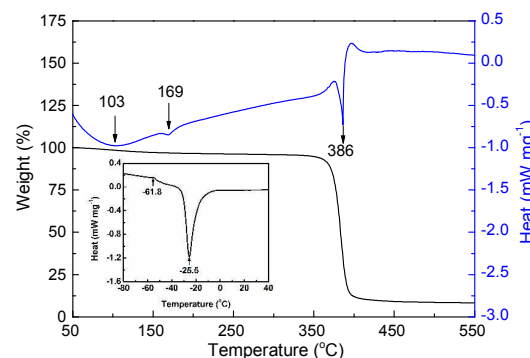
Samples	$A(S\text{ cm}^{-1}\text{ K}^{0.5})$	$B(K)$	$T_0(^{\circ}\text{C})$	$E_a(\text{kJ mol}^{-1})$
B-PEG/LiTFSI	1.21	670.05	-63.85	5.57
B-PEG/PVDF/LiTFSI	0.93	646.54	-61.71	5.38

An efficient electrolyte must have an electrochemical window larger than the operating voltage of cathode. To measure the electrochemical stability of B-PEG/PVDF/LiTFSI, linear sweep voltammetry tests were applied and the results are shown in Fig. 3. With increasing temperature, the electrochemical decomposition potential is 5.23, 5.05, 4.88 and 4.81 V respectively at 25, 80, 120 and 140 $^{\circ}\text{C}$. Even at 140 $^{\circ}\text{C}$, the electrochemical decomposition potential of the electrolyte still exceeds the value that the LiFePO_4 cathode requires.⁴³ However, with the increase in temperature, the decomposition potential of SPE gradually decreases. This decrease may relate to the easier oxidation of electrolyte at higher temperature, which is common for most electrolytes, such as liquid electrolyte and SPE.⁴⁴

**Figure 3** The linear sweep voltammetry results of the B-PEG/PVDF/LiTFSI electrolyte membranes at 25, 80, 120 and 140 $^{\circ}\text{C}$.

Besides, the thermal stability of the electrolyte membrane was investigated. Fig. 4 presents the TG-DSC results of B-PEG/PVDF/LiTFSI. Only one weight loss in the temperature range of 364 - 396 $^{\circ}\text{C}$ is observed, indicating that the B-PEG/PVDF/LiTFSI is thermally stable below 364 $^{\circ}\text{C}$. For the DSC curve, three endothermic peaks appear along with the rising temperature. A wide peak around 103 $^{\circ}\text{C}$ would correspond to the volatilization of water absorbed during the process of the test sample preparation for the TG-DSC measurement, with less than 3% of weight loss. A tiny peak at 169 $^{\circ}\text{C}$ may be due to the melting of PVDF.⁴⁵ The sharp peak at 386 $^{\circ}\text{C}$ is assigned to the decomposition of the electrolyte membrane, showing

high thermal stability of the electrolyte membrane that will be excellent for manufacture of high safety lithium ion batteries.

**Figure 4** The TG-DSC results of the B-PEG/PVDF/LiTFSI electrolyte membrane. Inset: The DSC result of B-PEG polymer.

The neat polymer's glass transition temperature (T_g) was also determined by the DSC test (see the inset of Fig. 4). As seen in the heating process, the endothermic peak appearing at -61.8 $^{\circ}\text{C}$ is signature of the glass-amorphous transition and corresponds to the T_g . A cold crystallization is observed at -25.5 $^{\circ}\text{C}$, which is similar to the previous study.⁴⁶

3.3 Wide temperature performance of the triPEG-boron based electrolyte $\text{LiFePO}_4/\text{Li}$ battery

The B-PEG/PVDF/LiTFSI electrolyte membranes were assembled into the $\text{LiFePO}_4/\text{Li}$ batteries. The cycling performance at 80 $^{\circ}\text{C}/1\text{ C}$ and 25 $^{\circ}\text{C}/0.3\text{ C}$ is shown in Fig. 5a, and the AC impedances corresponding to the cycle tests are shown in Fig. 5b. The specific capacity of 103 mAh g^{-1} at 80 $^{\circ}\text{C}$ and 1 C in the first cycle is obtained. As the cycles proceed, the capacity gradually increases until the 125 mAh g^{-1} capacity is stabilized at the 10th cycle. It is notable to mention that the capacity of 125 mAh g^{-1} is maintained for 200 cycles, showing stable cycling performance of the battery.

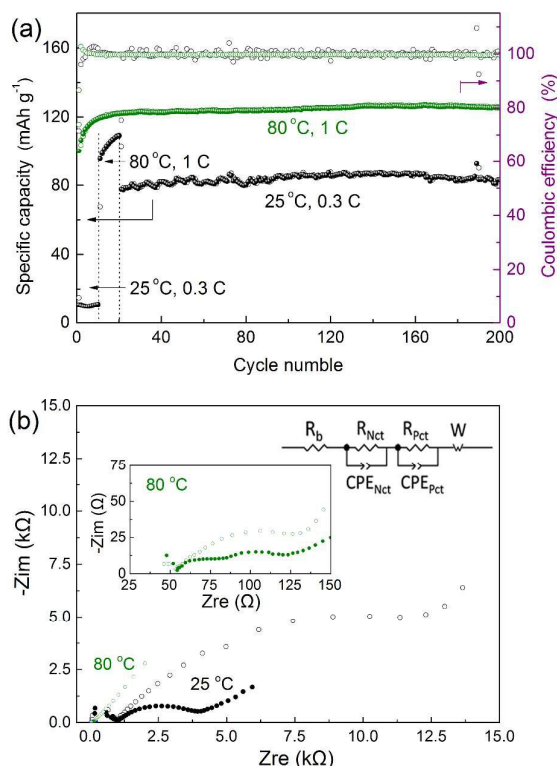


Figure 5 (a) The cycling performance of the LiFePO₄/B-PEG/PVDF/LiTFSI/Li batteries at 80 °C/1 C (olive), and 25 °C/0.3 C (black) with a pre-activation process of 80 °C/1 C. (b) The AC impedances at selected conditions and the equivalent circuits, filled dots represent the beginning of the first cycle, hollow dots are the end of the 20th cycle.

At 25 °C/0.3 C, however, there is a low capacity of 10 mAh g⁻¹ in the initial cycle. In order to solve this problem, the AC impedance spectroscopy was used to trace the resistance change of the battery. It is found in Fig. 5b that the interfacial resistance is obviously decreased as the temperature is increased from 25 to 80 °C. Therefore, a high-temperature pre-activation process at 80 °C and 1 C for 10 cycles was carried out before the cycling measurements at room temperature. A capacity of 82 ± 4 mAh g⁻¹ is obtained at 25 °C/0.3 C during the 200 cycles. What is more, the capacity retention rate is nearly 100% (note: the fluctuations of the capacities were caused by the temperature fluctuations in the test room).

An obvious capacity rising can be seen after the initial 10 cycles. This rising period is induced by the reduction of battery resistance. In Fig. 5b, the cell resistance consists of four parts: the intersection of the high frequency area semicircle and the x axis represents the ionic conduction resistance in the bulk of the cell (R_b), the high frequency area semicircle represents the charge transfer resistance at the Li/electrolyte interface (R_{Nct}), the intermediate frequency area semicircle represents the charge transfer resistance at the LiFePO₄/electrolyte interface (R_{Pct}) and the low frequency area straight line represents the diffusion resistance of lithium ions in LiFePO₄ and the electrolyte (W). Additionally, the CPE_{Nct} and CPE_{Pct} represent the capacitive reactance at the Li/electrolyte interface and LiFePO₄/electrolyte interface respectively.

The R_b does not change at varying temperatures during the cycles, illustrating high stability of the electrolyte. However, the interface resistance changes significantly. At 25 °C the sum of R_{Nct} and R_{Pct} decreases from 8145 to 2112 Ω, which is a reduction of 74%, and at 80 °C is decreased from 61 to 52 Ω with 15% dropped. This illustrates the point-plane and point-point contacts at the electrolyte/electrode interfaces are the main issue. Thus, a high-temperature pre-activation process is helpful for improving the low temperature performance of the all-solid-state battery.

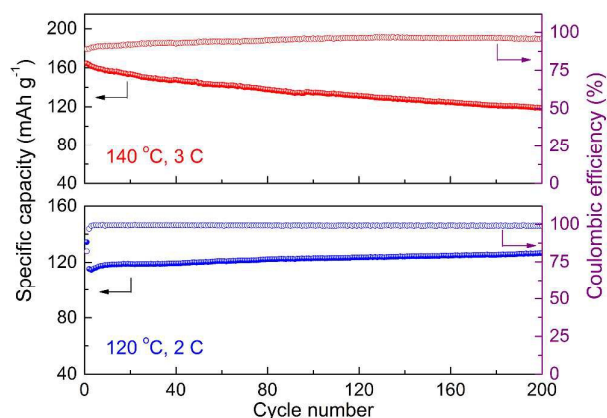


Figure 6 The high temperature cycling performance of the LiFePO₄/B-PEG/PVDF/LiTFSI/Li batteries at 120 °C/2 C and 140 °C/3 C.

At 120 °C/2 C, the initial capacity is 111 mAh g⁻¹, and the average capacity for the 200 cycles is 123 mAh g⁻¹ with a coulombic efficiency of 99%. When a higher temperature of 140 °C and a larger rate of 3 C were applied, the initial discharge capacity reaches 146 mAh g⁻¹, and the capacity is 113 mAh g⁻¹ after the 200 cycles, in which the coulombic efficiency is about 87% for the first cycle and 95% for the 200th cycle.

4. Conclusions

The triPEG-boron based electrolyte membrane was fabricated, displaying excellent thermal and conductivity properties. The assembled LiFePO₄/Li batteries with the electrolyte membranes show the wide operating temperature range of 25 - 140 °C. This greatly broadens the application range of all-solid-state lithium ion battery. Moreover, the filming method for the electrolyte is simple and efficient and is advantageous in the fabrication of all-solid-state lithium ion battery.

Acknowledgements

This work was supported by the National Natural Science Foundation of China (No. 51274239) and Central South University, which are greatly appreciated.

Notes and references

ARTICLE

Journal Name

- 1 D. Zhou, Y.B. He, R. Liu, M. Liu, H. Du, B. Li, Q. Cai, Q.H. Yang and F. Kang, *Adv. Energy Mater.*, 2015, **5**, 1500353.
- 2 Q.J. Wang, W.L. Song, L.Z. Fan and Q. Shi, *J. Power Sources*, 2015, **279**, 405-412.
- 3 J. Shim, D.G. Kim, H.J. Kim, J.H. Lee, J.H. Baik and J.C. Lee, *J. Mater. Chem. A*, 2014, **2**, 13873-13883.
- 4 R. Khurana, J.L. Schaefer, L.A. Archer and G.W. Coates, *J. Am. Chem. Soc.*, 2014, **136**, 7395-7402.
- 5 J.F.M. Oudenhoven, L. Baggetto and P.H.L. Notten, *Adv. Energy Mater.*, 2011, **1**, 10-33.
- 6 M. Patel, M.U.M. Patel and A.J. Bhattacharyya, *ChemSusChem*, 2010, **3**, 1371-1374.
- 7 C. Zhang, Y. Lin and J. Liu, *J. Mater. Chem. A*, 2015, **3**, 10760-10766.
- 8 D. He, D.W. Kim, J.S. Park, S.Y. Cho and Y.K. Kang, *J. Power Sources*, 2013, **244**, 170-176.
- 9 E. Quartarone and P. Mustarelli, *Chem. Soc. Rev.*, 2011, **40**, 2525-2040.
- 10 D.E. Fenton, J.M. Parker and P.V. Wright, *Polymer*, 1973, **14**, 589-593.
- 11 B.W. Zewde, S. Admassie, J. Zimmermann, C.S. Isfort, B. Scrosati and J. Hassoun, *ChemSusChem*, 2013, **6**, 1400-1405.
- 12 J. Shim, D.G. Kim, J.H. Lee, J.H. Baik and J.C. Lee, *Polym. Chem.*, 2014, **5**, 3432-3442.
- 13 Y. Wang and W.H. Zhong, *ChemElectroChem*, 2015, **2**, 22-36.
- 14 N.P. Young, D. Devaux, R. Khurana, G.W. Coates and N.P. Balsara, *Solid State Ionics*, 2014, **263**, 87-94.
- 15 J. Li, Y. Lin, H.H. Yao, C.F. Yuan and J. Liu, *ChemSusChem*, 2014, **7**, 1901-1908.
- 16 C.Y. Tang, K. Hackenberg, Q. Fu, P.M. Ajayan and H. Ardebili, *NanoLett.*, 2012, **12**, 1152-1156.
- 17 S. Lee, M. Schomer, H. Peng, K.A. Page, D. Wilms, H. Frey, C.L. Soles and D.Y. Yoon, *Chem. Mater.*, 2011, **23**, 2685-2688.
- 18 Y. Lin, J. Li, Y.Q. Lai, C.F. Yuan, Y. Cheng and J. Liu, *RSC Adv.*, 2013, **3**, 10722-10730.
- 19 Y.F. Tong, L. Chen, X.H. He and Y.W. Chen, *Electrochim. Acta*, 2014, **118**, 33-40.
- 20 C. Gerbaldi, J.R. Nair, M.A. Kulandainathan, R.S. Kumar, C. Ferrara, P. Mustarelli and A.M. Stephan, *J. Mater. Chem. A*, 2014, **2**, 9948-9954.
- 21 K. Zhu, Y.X. Liu and J. Liu, *RSC Adv.*, 2014, **4**, 42278-42284.
- 22 J.H. Choi, S.H. Kim, H.J. Ha, E.H. Kil, C.K. Lee, S.B. Lee, J.K. Shim and S.Y. Lee, *J. Mater. Chem. A*, 2013, **1**, 5224-5231.
- 23 J. Rolland, J. Brassinne, J.P. Bourgeois, E. Poqqi, A. Vlad and J.F. Gohy, *J. Mater. Chem. A*, 2014, **2**, 11839-11846.
- 24 N.S. Choi, Y.M. Lee, K.Y. Cho, D.H. Ko and J.K. Park, *Electrochem. Commun.*, 2004, **6**, 1238-1242.
- 25 Y. Kato, S. Yokoyama, H. Ikuta, Y. Uchimoto and M. Wakihara, *Electrochem. Commun.*, 2001, **3**, 128-130.
- 26 H.S. Lee, X.Q. Yang, C.L. Xiang, J. McBreen, L.S. Choi, *J. Electrochem. Soc.*, 1998, **145**, 2813-2818.
- 27 K.L. Mathews, A.M. Budgin, S. Beeram, A.T. Joenathan, B.D. Stein, U. Werner-Zwanziger, M. Pink, L.A. Baker, W.E. Mahmoud, J.P. Carini and L.M. Bronstein, *J. Mater. Chem. A*, 2013, **1**, 1108-1116.
- 28 Y.M. Lee, J.E. Seo, N.S. Choi and J.K. Park, *Electrochim. Acta*, 2005, **50**, 2843-2848.
- 29 J. Shim, D.G. Kim, H.J. Kim, J.H. Lee and J.C. Lee, *ACS Appl. Mater. Interfaces*, 2015, **7**, 7690-7701.
- 30 F. Kaneko, S. Wada, M. Nakayama, M. Wakihara, J. Koki and S. Kuroki, *Adv. Funct. Mater.*, 2009, **19**, 918-925.
- 31 H. Aydın and A. Bozkurt, *J. Appl. Polym. Sci.*, 2012, **124**, 1193-1199.
- 32 A. Sezgin, U. Akbey, M.R. Hansen, R. Graf, A. Bozkurt and A. Baykal, *Polymer*, 2008, **49**, 3859-3864.
- 33 H. Aydın, M. Şenela, H. Erdemib, A. Baykala, M. Tülüa, A. Atac and A. Bozkurt, *J. Power Sources*, 2011, **196**, 1425-1432.
- 34 S. Heřmánek, *Chem. Rev.*, 1992, **92**, 325-362.
- 35 P.J. Bray, *Inorg. Chim. Acta*, 1999, **289**, 158-173.
- 36 H. Wu, O.T. Cummings and C.D. Wick, *J. Phys. Chem. B*, 2012, **116**, 14922-14932.
- 37 C.F. Yuan, J. Li, P.F. Han, Y.Q. Lai, Z. Zhang and J. Liu, *J. Power Sources*, 2013, **240**, 653-658.
- 38 J.M. Tarascon and M. Armand, *Nature*, 2001, **414**, 359-367.
- 39 Y. Kumar, G.P. Pandey and S.A. Hashmi, *J. Phys. Chem. C*, 2012, **116**, 26118-26127.
- 40 Z.C. Zhang, J.J. Jin, F. Bautista, L.J. Lyons, N. Shariatzadeh, D. Sherlock, K. Amine and R. West, *Solid State Ionics*, 2004, **170**, 233-238.
- 41 R. Yuan, A.A. Teran, I. Gurevitch, S. A. Mullin, N.S. Wanakule and N.P. Balsara, *Macromolecules*, 2013, **46**, 914-921.
- 42 Z.G. Xue, D. He and X.L. Xie, *J. Mater. Chem. A*, 2015, **3**, 19218-19253.
- 43 F. Omenya, B. Wen, J. Fang, R. Zhang, Q. Wang, N.A. Chernova, J. Schneider-Haefner, F. Cosandey and M.S. Whittingham, *Adv. Energy Mater.*, 2015, **5**, 1401204.
- 44 K. Xu, *Chemical reviews*, 2004, **104**, 4303-4418.
- 45 C.L. Yang, Z.H. Li, W.J. Li, H.Y. Liu, Q.Z. Xiao, G.T. Lei and Y.H. Ding, *J. Membrane Sci.*, 2015, **495**, 341-350.
- 46 P.Y. Pennarun and P. Jannasch, *Solid State Ionics*, 2005, **176**, 1103-1112.

A wide operating temperature electrolyte membrane is fabricated for all-solid-state lithium ion battery.

